DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics, and Computer Science

Greedy TA Review *Triathlon*

CSE2310 Algorithm Design

Report

For this exercise, we expect you to write a report in **your own words**. Your report should meet the following constraints:

- It should be a **PDF** file with at most two sides of an A4 (excluding an optional front page or space used by diagrams). The use of LaTeX is recommended.
- It should be written in clear, readable, and correct English.
- It should contain for each question:
  - a clear and brief explanation (intuition) about why your answer/(pseudo)code is correct,
  - an indication where you have doubts or are stuck.

General recommendations and hints

- Read the complete assignment carefully before you start solving the subproblems.
- If you get stuck: don't panic, but read the relevant sections in the book first. You can also ask the assistants to help you.

You are asked to compute in which order n contestants of a triathlon should be sent on their way such as to minimize the (expected) *completion time*, which is the earliest moment everyone is finished. The triathlon consists of first swimming in the pool, then biking a certain distance, and finally running a small distance. Biking and running can be done simultaneously, but the pool can be used only by one contestant at a time.

From previous competitions, there is data on the expected times for each of the contestants and for each of the three activities: for $i = 1, \ldots, n$, let w_i, b_i, and r_i be the times contestant i requires for swimming, biking and running, respectively.

1. Suppose we have two contestants, one with $w_1 = 6$, $b_1 = 9$, and $r_1 = 6$ and the other with $w_2 = 9$, $b_2 = 4$, and $r_2 = 7$. What is the optimal order of these contestants (minimizing the completion time) and what is this completion time?
2. Give pseudocode of a greedy algorithm to determine a schedule that minimizes the completion time (in at most 10 lines of code). Let the algorithm compute all start times $s_1, s_2, \ldots, s_n$.
3. Give the tightest worst-case upper bound on the running time of your algorithm.
4. Prove that your algorithm from the previous question returns a schedule $S = \{s_1, s_2, \ldots, s_n\}$ with a completion time that is not later than an optimal schedule (in at most 25 lines). Use an exchange argument for this proof. (You may assume your algorithm is correct.)